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FEATURED ARTICLE

Brain-Computer Interfaces

ROHAN SELVA-RADOV

editor's note

It's amazing to see how far this society and journal have come; the 3rd issue of HYPERTEXT shows how much persistence we as Olavian computer science students have.

HYPERTEXT has not only survived through these stressful times of COVID, but in fact flourished; we have a huge variety of interesting articles in this edition, ranging from Artificial General Intelligence to Machine Learning and the crossover between the fields of music and tech.

Issue 3 clearly shows the sheer enthusiasm of these writers; putting time aside from what must be very busy schedules and workloads to research a topic of interest and write an article is an impressive feat; thank you to all those who have written an article and allowed HYPERTEXT to continue.

Feedback is welcomed! Please contact your ocomputer science teacher, the Head of Computing at rzeshan@saintolaves.net or myself at hardiv.harshakumar@saintolaves.net. I look forward to taking a look at any suggestions you have or any questions that need answering.

Thanks once again to everyone who submitted an article, and I hope you enjoy this issue of HYPERTEXT.

Don't forget to #bleedpurple, and Happy Reading!

HARDIV HARSHAKUMAR

~ CHIEF EDITOR OF HYPERTEXT, PRESIDENT OF COMPUTER SCIENCE SOCIETY

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HAPPY READING!

Brain-Computer Interfaces

Rohan Selva-Radov

If you want to predict the future, there are worse places to start than in your [local library's sci-fi section](#). The internet, the Apollo 11 moon lander, and mobile phones are just a few inventions predicted by science fiction writers decades – and in some cases centuries – before their creation in the real world. But one futuristic fixture of such novels is still notably absent from our world: brain-computer interfaces (BCIs), devices which enable direct two-way communication between our brains and computers. Arguably first envisioned by E. V. Odle in his 1923 cyborg classic “The Clockwork Man” (Newitz, 2021), BCIs would take the digitalisation of our day-to-day lives a step further, by enabling communication between mind and machine via the medium of wires, rather than through the filter of one of our five senses (Parliamentary Office of Science and Technology, 2020).

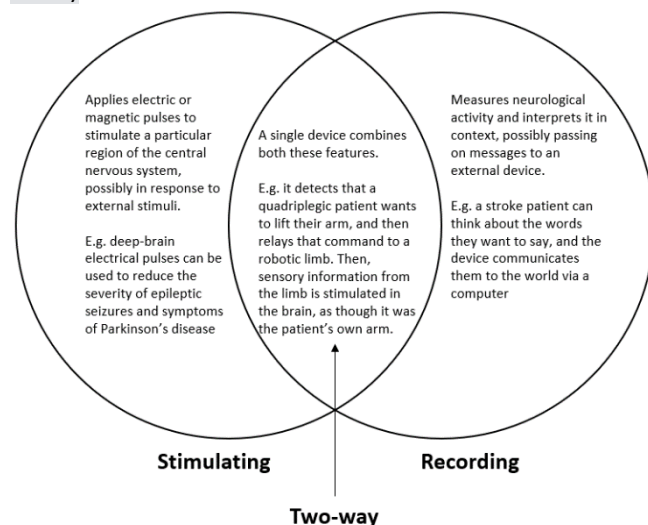


Figure 1, the different classifications of BCI devices and illustrative medical applications of each. Source, (Parliamentary Office of Science and Technology, 2020)

Before everyone has a “Fitbit in [their] skull”, in the words of Elon Musk, neural implants (NIs) must overcome two distinct hurdles. First, the formidable practical challenge of how to capture our brain waves and translate them into a computer-readable format. Second, and less obviously, the difficult goal of

convincing regulators, politicians, and the public that the technological benefits of BCIs outweigh the myriad of potential societal risks.

On the physical side, there is a fundamental trade-off between resolution and intrusion which must be struck: to better detect precisely which neurons are firing at a given moment, the BCI sensor must be more closely integrated into the brain (Parliamentary Office of Science and Technology, 2020).

At the least invasive side of the spectrum, an electroencephalogram (EEG) or magnetoencephalogram (MEG) may be used to allow doctors to measure brain activity without needing any surgical intervention – devices not dissimilar to the cap of electrodes worn by Eleven whilst at the Hawkins Lab, in *Stranger Things*. To unlock more detail about what is going on in the brain, chips must be directly implanted into your grey matter, and communicate wirelessly with the outside world. This is the approach taken by Neuralink, who last year showcased a pig which had had a thousand ultra-fine electrodes injected into its brain (Regalado, 2020).

This acquisition of neurological information is only the first stage of a BCI, though. Particularly for non-invasive approaches, raw data have extremely limited utility. The fact that these electric or magnetic waves must pass through the skull means that the signal picked up by these devices is extremely “noisy”, particularly when the user is going about their day-to-day business. This is compounded by the fact that listening in to the action from afar means that it’s extremely difficult to pinpoint which neurons are firing at a given time, and thus hard to work out exactly what the patient’s brain signals mean. The next stages, of signal amplification, noise reduction, and feature generation, are just as important as collecting the primary data, and every step requires finely tuned algorithms to extract useful patterns and features (Bashashati, Fatourechi, & Ward, 2007).

Developments in machine learning have made it easier for researchers to personalise these signal-cleaning algorithms to each user, as well allowing them to combine simultaneous EEG readings from across the skull, and ultimately build a more spatially accurate picture of neural activity. This is something which is vital to determining the user’s intent, given that the different cortexes, lobes, and regions of the brain are all responsible for different activities (Durham, 2019).

As well as improving the software used to process brain signals, improvements can also be made to the hardware. Over the past three years though, DARPA,

the USA's military research agency, has ploughed millions of dollars into "minimally invasive" BCI devices which would improve signal quality at the source, rather than processing it post-collection. One team has proposed using a viral vector to edit the DNA of a few hundred neurons in brain regions of interest, so that they emit infrared light when stimulated, a form of EM radiation which is not distorted by passage through the skull.

DARPA hopes BCIs could have military applications as widespread as allowing a general to pilot information from a swarm of drones simultaneously, to mitigating the effects of PTSD (DARPA, 2018). And, if these projects come to fruition, there could be an explosion in the use of BCIs in able-bodied people: DARPA-funded innovations have a tendency to trickle down into civilian use – the internet, for instance, started life as a DARPA project. Whilst our physical health can be analysed by heart monitors and the like, there are comparatively few similar devices which can track our mental wellbeing. It's easy to see how appealing a small, portable device that provides quantitative insights into our mood; tips about our learning style; or early advice about our risk of developing mental illnesses like dementia or depression, might be (Rosso, 2019).

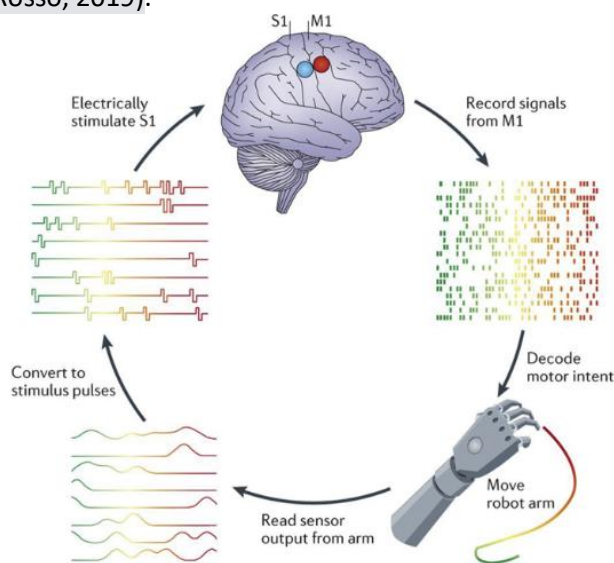


Figure 2: an illustration of how a two-way NI could control a robotic arm and receive sensory feedback. (Bensmaia & Miller, 2014)

So far, all the applications discussed have involved taking neurological information and transmitting it to computers. But there is a reverse of this too: taking computational information and transmitting it into your mind. The idea of a computer controlling your brain may sound rather Matrix-esque, but medical applications of this sort of technology have been around since the 1970s, in the form of the cochlear implants commonly used to treat acute hearing loss.

These small devices pick up and process sounds from the environment, feeding them in directly to the user's auditory nerve as electrical impulses. This process bypasses the patient's own, faulty biological analogue-to-digital converters in the inner ear, and allows them the experience of hearing normally (Eshraghi, et al., 2012). Though remarkable, this is small fry compared with what the likes of Neuralink hope to achieve. Rather than merely enabling the disabled or infirm to live more able-bodied lives, the companies at the forefront of BCI innovation envisage integrating our biological intelligence with the artificial kind more closely than ever before.

Mark Zuckerberg recently set out his vision for Facebook to create a "metaverse", a shared digital universe where the physical and online worlds interact through virtual reality and augmented reality. Facebook's funding of research into mind-reading headsets that translate your thoughts into text is a central plank of this ambition, aiming to make interacting with this metaverse as seamless as possible (Bradshaw & Murphy, 2021). Elon Musk's company Neuralink, meanwhile, has been explicit in its long-term mission of developing a device that will enable healthy humans to be enhanced in new ways, something known as transhumanism (Neuralink, 2021). This mingling of carbon and silicon will, hope the tech giants, enable people to spend much of their lives in an all-encompassing augmented reality world. Neuralink and the others on the invasive side of BCI technology are betting that the appeal of joining this vibrant new binary universe will be sufficient to persuade people that they really do want a robotic sewing machine plunging thousands of metal wires into their head.

But, for all the excitement of the early adopters, BCI's mass acceptance face a formidable obstacle: the mountain of ethical qualms and concerns which is bound to come with technological intrusion into our minds themselves. These fears will be particularly acute for invasive BCIs, as they cannot be disabled simply by pulling off a cap of electrodes, like with non-invasive devices. Whilst concerns that they might "read our mind" are overblown for the moment (as users often have to consciously make the BCI aware of each thought, like how we must actively transfer our mental ideas into words), the pace of development in the sector means that BCIs implanted today could, in time, have software upgrades that do allow them to passively track our thoughts (Caterina, Davide, & Riccardo, 2019). If, or maybe when, this mind reading phase arrives, who would have the ownership of any insights thereby generated? If you had a Facebook BCI, would you be content with them using data they

extract from your brain to better target adverts? Would you feel comfortable about having a Google-produced BCI chip in your brain which can stimulate emotions and sentiments? The general public apathy towards the constant tracking we face online may translate into an indifference about Big Tech muscling their way into our minds, cementing their oligopolies in the digital world.

If the thought of Western business giants nosing around your thoughts is concerning, then the threat of state surveillance may be still more worrying. The ability to monitor thoughts and sentiments would strengthen China's already-sprawling surveillance state to the heights of intrusion outlined by George Orwell in his dystopian masterpiece "1984" (Qiang, 2019). And there is, of course, the threat of hacking by opportunistic criminals: how many Bitcoins would you be willing to pay to free your BCI of ransomware? At a more philosophical level, policymakers must also consider where responsibility lies when BCIs are involved.

There are some parallels here with the regulation of self-driving cars, but the issue becomes particularly challenging when it comes to the sort of mind-to-mind communications being funded by DARPA. The agency is currently exploring technologies that would enable one soldier to share detailed information about the terrain to another, straight from the brain of the first to the second (Boyd, 2021). Imagine that an infantrywoman on the front line wears a camera and receives neural, live updates from a remote intelligence officer monitoring the video. Who would be held accountable – if anyone at all – if the AI software processing the intelligence officer's brain signals deemed they had identified a potential risk, directed the infantrywoman to fire, and ultimately resulted in civilian deaths?

The practical consequences of all these questions may be far-off into the future, but scientific progress tends to come slowly, then suddenly. Failing to keep developments in BCIs on the right side of the public's moral boundaries will result in widespread opposition to the entire industry, robbing ourselves of the chance to reap the benefits of this truly ground-breaking technology (Trimper, Wolpe, & Rommelfanger, 2014). Some might argue that it is futile to speculate about how BCIs may develop: it is difficult to know for certain whether they will become commonplace, and impossible to consider the "unknown unknowns" about their application. Despite this, forward-thinking consideration of the potential issues is paramount. Here, lessons must be learned from the advent of the Internet – in its early days, academics and researchers

envisaged a very different web to the one we see today, both in terms of promise and pitfalls. Rather than naively assuming that our regulators and policymakers will adapt as BCIs develop, we should pre-empt the most obvious ethical challenges, like the ones mentioned above, as well as creating institutions which are sufficiently flexible to react to unforeseen issues. The Royal Society's 2019 listening exercise (van Mil, Hopkins, & Kinsella), which sought to gauge the views of a cross-section of society on BCIs, was a welcome start, but much more public consultation is needed.

The past five years has seen an explosion in the number of firms, plucky startups and corporate behemoths alike, interested in the immense potential of BCIs. From Facebook to Neuralink, firms are pushing the boundaries of engineering and ethics, and further blurring the line between neuro- and computer science. Perhaps those 20th-century sci-fi authors were not so fanciful after all.

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Genetic Algorithms

Grace Sodunke

What are Genetic Algorithms?

Finding an optimal solution to a problem with constraints has inspired the development of numerous techniques in computer science, mathematics and engineering. All feasible solutions are part of a finite set of discrete items from which the best element is selected. Consequently, heuristic algorithms that reduce the search space provide a vastly more efficient alternative than exhaustive search. When modelled as decision problems, these problems are NP-complete meaning that there is no algorithm that efficiently solves for large input.

A subset of heuristic algorithms is evolutionary algorithms (EAs), which are based on the principle of Darwinian evolution. Biological species have taken advantage of chaos and chance to produce optimal characteristics; a consequence of natural selection is that organisms most capable of survival will become prevalent within a species. As a result, EAs are of interest for solving optimisation problems since we can undoubtedly arrive at a good solution by making a series of small modifications to an object.

Among evolutionary techniques, genetic algorithms (GAs) are the most extensively applied. A genetic algorithm is a stochastic search algorithm that iteratively adapts a population of individuals to create better generations. The suitability of individuals as a solution is evaluated using a fitness function – this is the objective function used to guide the search. In each iteration, successive solutions are created by combining two parents rather than modifying a single individual. This distinguishes GAs from traditional search algorithms like gradient descent which check one solution at a time. By adapting a large population of individuals, an efficient search towards the optimal solution is performed.

Consider the problem of maximising the function

$$f(x) = -x^2 + 10x - 3$$

for

$$0 \leq x \leq 31$$

, which is plotted in Figure 1.

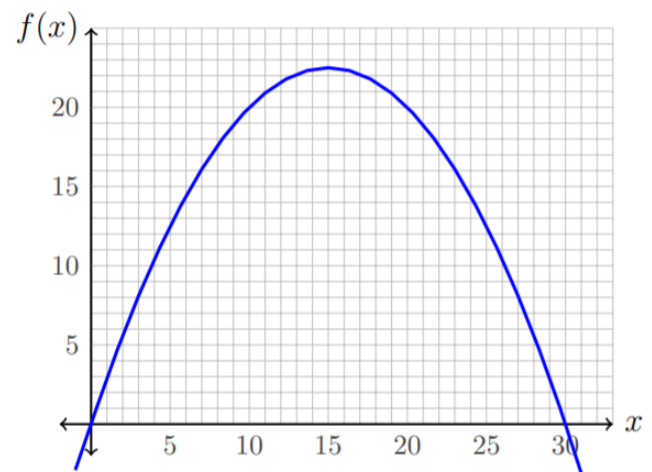


Figure 1

The first step is to randomly initialise a population of candidate solutions. These are called chromosomes and are typically encoded as a bit vector. Other implementations can use arrays of numbers or another object that is appropriate to the problem. In this example we will encode

xx

as a binary integer of length 5 so that

xx

ranges from 0 (00000) to 31 (11111). Furthermore, the population size is a parameter that is determined from the complexity of the problem; this example will use a population size of 10.

Next, a fitness function is applied to each individual to return a fitness value. A higher value is returned for better solutions as it gives them a greater chance to propagate. The fitness values for the initial population are given in Table 1.

Chromosome number	Initial Population	xx	Fitness value $f(x)$	Selection probability
1	01011	11	20.9	0.1416
2	11010	26	10.4	0.0705
3	00010	2	5.6	0.0379
4	01110	14	22.4	0.1518
5	01100	12	21.6	0.1463
6	11110	30	0	0
7	10110	22	17.6	0.1192
8	01001	9	18.9	0.1280
9	00011	3	8.1	0.0549
10	10001	17	22.1	0.1497

Table 1

The next step is to select two parents from the population. In roulette wheel selection, the probability of selecting an individual is proportional to its fitness. Therefore

$$P(\text{choosing chromosome } i) = \frac{f(i)}{\sum_{j=1}^{10} f(j)}$$

Alternatively, tournament selection involves choosing

mm
 individuals randomly (where
 mm
 is the tournament size) and selecting the winner with the highest fitness.
 When two parents are selected, they are combined to produce a stronger solution in the crossover stage. A crossover point is randomly selected in the two strings, and all values before this position are swapped produce two child solutions. This operation is a primary advantage of the algorithm.



Figure 2

After crossover, the children are subject to mutation. Mutations prevent the algorithm from converging to a local optimum, since diversity is maintained in the population of solutions. This ensures the search space is ergodic since any solution can be generated with a non-zero probability. Without this step, the quality of the final solution is always limited by genes in the initial population. Mutation of a bit involves flipping its value according to a small independent probability. The child solutions are added to a new population. When the population size is reached, a new generation is produced and the algorithm executes another iteration. The search terminates according to any of the following stopping conditions:

- The maximum number of generations is reached
- A specified time has elapsed
- The maximum fitness remains constant after several generations.

Table 2 shows the solutions produced in the second generation. The average fitness has increased from 14.76 to 17.48, illustrating how the algorithm ‘evolves’ towards better solutions.

Chromosome number	Parents	Children	Xx	Fitness value $f(x)f(x)$
5	01 100	01010	10	20
2	11 010	11100	28	5.6
4	0111 0	01111	15	22.5
8	0100 1	01000	8	17.6
9	0001 1	01010	10	20
2	1101 0	11011	27	8.1
7	1011 0	10110	22	17.6
4	0111 0	01110	14	22.4
10	100 01	10001	17	22.1
8	010 01	01001	9	18.9

Table 2 (mutations shown in bold)

A genetic algorithm involves random sampling, so why is it guaranteed to converge to an optimal solution? A theorem proposed by John Holland in the 1970s

explains this using the idea of a schema. An example of a schema is 10*1* which represents the set of strings with ‘1’s in position 1 and 4 and ‘0’ in position 2. The ‘*’ is a wildcard operator and can be either 1 or 0. The order of a schema is the number of fixed bits in the string (i.e. not ‘*’) while the defining length is the distance between the first and last fixed bits. Holland’s schema theorem states that low-order schema with a short defining length and above-average fitness are more likely to appear in successive generations. Hence good schemata will grow rapidly in the population.

Genetic algorithms are widely employed as heuristics in combinatorial problems. In the travelling salesman problem (TSP), n cities are to be visited each once before returning to the start city, and the shortest route must be found. The fitness function is defined as the cost of a particular route, and the crossover operation is modified to add cities from two parents in a cyclic order. In the knapsack problem, a set of n items each have a weight and a value; given a knapsack of fixed capacity, the items put inside should have the largest total value possible. To implement a GA, each chromosome is a string of bits with ‘1’ showing that an item is included and ‘0’ if not.

In several fields of engineering, genetic algorithms are used in manufacturing, design and routing. Certain problems are not solvable using gradient-based methods if it uses discrete variables or non-differentiable objective functions. Using a GA, the optimal dimensions of a component can be determined to maximise performance. Designing a processor will also require hardware security and power consumption to be prioritised. In network design, the fitness function will consider reliability and latency when creating topologies.

In machine learning, GAs are used to optimise weights in a neural network. It is also applied in feature selection of variables in a dataset, which can largely impact the performance of a classifier model. Reinforcement learning involves finding a policy that maximises a reward function, which can be modelled as a stochastic optimisation problem. Therefore GAs can be used in RL to optimise parameters in policy gradient methods.

The ability of GAs to consistently produce high-quality solutions powerfully demonstrates the benefits of seeking solutions to theoretical problems from natural processes.

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What happened to Intel?

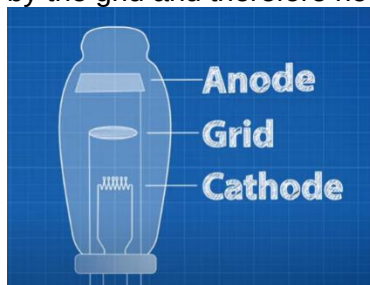
Hardiv Harshakumar

Intel is arguably one of the most renowned names in the semiconductor industry; as the CPU inventors, they were the go-to processor company in almost every desktop product. The introduction of the CPU revolutionised the world as we know it. In the 1950s and 60s, computers required specialised circuit chips for each application a computer performed. The CPU was a simple chip that could be programmed to run any application performed by these computer chips. The CPU, however, would not have been made possible if it were not for the transistor.

The transistor was invented at Bell Laboratories in 1947; its original purpose was simply to amplify electrical signals. Transistors can essentially behave as a switch, controlling the flow of electrons by being in on/off states. These on or off states could be represented as binary 0s and 1s, the foundation of modern computing as we know it. The predecessor to the transistor was the vacuum tube.

Vacuum tubes were evacuated glass bulbs that contained three main components: an anode, cathode and grid, each with its terminals. The concept is that electrons can flow from the cathode to the anode without much resistance due to the evacuated environment within the tube.

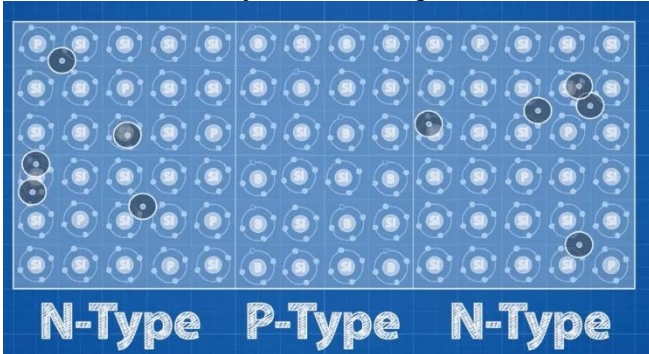
The function of the grid would be to change the output of the vacuum tube. Electrons would flow, as stated before if the grid were positively charged from the cathode to the anode. If it were negatively charged, electrons would be repelled by the grid and therefore not reach the anode.



This simple idea is what laid the foundation for binary coding - the fact that applying a current to a component could cause current to flow in another circuit meant that these switching could be represented by on and off signals - binary zeros and ones. However, it was clear that vacuum tubes were not sustainable in the long term. The world's first computer, the ENIAC, used around 18000 vacuum tubes to perform its calculations. This computer weighed almost 30 tonnes, took up the space of a room, and had a high power demand, all of these thanks to the use of vacuum tube circuitry. Furthermore, a lot of the vacuum tubes would end up burning out due to the heat applied to them, causing high maintenance costs as vacuum tubes were replaced regularly. Computers that utilised vacuum tube circuitry are called first-generation computers.

Transistors were the solution to this issue. Transistors utilised a much more microscopic design that harnessed the properties of semiconductor materials rather than relying on specific environmental conditions to allow a current to flow. Silicon is usually the material that is used for semiconductor chips. Since it is a semiconductor, its conducting properties can be tailored by introducing impurities to the crystal structure, making it a suitable medium for controlling electrical current. Silicon has 4 electrons in its valence shell and so forms covalent bonds with four neighbouring silicon atoms to gain those extra electrons. If we introduce those impurities to this pure silicon crystal, we can change how it conducts a current. Adding phosphorus (5 valence electrons in the outer shell) adds a free electron to the structure and causes the transistor to be negatively charged, hence the N-type semiconductor. P-type semiconductors are positively charged since its doped with Boron, which has 3 electrons in its valence shell. The Boron requires its final electron and uses electrons from its neighbours, creating a positive mobile charge called a hole. This allows for positive charges to move through p-type conductors. As the number of charges in a semiconductor increases, so will its conductivity. The most prevalent transistor is the NPN transistor. In an NPN transistor, Electrons and holes interact at the n-type p-type junction, creating a depletion layer. This prevents electrons from passing through the junction so that no current flows through the transistor. However, when a positive current is applied to the p-type section, the depletion layer no longer exists, and

electrons can freely move through the transistor.



The transistor offers many benefits over the vacuum tube - it is more reliable, cheap, and less susceptible to damage. Transistors also produce a lot less heat than vacuum tubes. The most notable, however, which simply cannot be emphasised enough - is the size of the component itself. Modern transistors can be as small as 5nm in length, compared to the most miniature vacuum tube with a diameter of 20.44mm. Size is a great advantage. The idea is that the shorter the connections between transistors in a computer, the more efficient and reliable the computer would be. When transistors were invented and started getting adopted into electronic devices and circuits, wires were the main limiting factor in terms of just how small circuits could get. A new way of arranging components in circuits was required, and subsequently, the method of producing PCBs was born. PCBs allowed for connections between components to be efficiently organised so that the distance electrons travelled was minimised. They also paved the way for the mass manufacture of electronics, as connections between components could simply be etched onto a board rather than being connected by humans by hand.

With these innovations came the Integrated Circuit (IC), which allowed multiple components to be integrated onto the same chip and paved the way for CPUs to be made. Integrated Circuits tightly packed transistors together but allowed them to function independently. The first CPU came from Intel, a startup born from the company known as Fairchild semiconductor. They were the ones that had produced a whole computer on a chip' - the computing power of the ENIAC could now all fit on a fingertip. Intel pushed further with the release of its 386 and Pentium chips which were also on the top of the market at their time. It seems as though Intel was at the top of the industry from the start and has remained so ever since. However, this success has not stayed around in recent years. Intel is the only chip designer that both owns and operates foundries to manufacture these chips. This means all design and manufacture of intel processors are

done in-house. This used to be suitable for intel in the early days, as it allowed for quick tweaks to the hardware and more customisation. However, this has now become a curse, and Intel has started to slip. It simply cannot keep up with other chip manufacturing companies such as tsmc and Samsung.

The reason for this? Both TSMC and Samsung have been able to achieve 7nm processes for their CPU chips. A 7nm chip means that more transistors can be tightly packed together in a given area, which significantly increases the efficiency and speed of a processor whilst also reducing its thermal output. A 7nm fabrication process can offer up to 25% more performance than a 14nm process. This is the main reason intel have been struggling with their current processors - Intel had planned to make the jump from a 14nm fabrication process to a 10nm process in 2015 and still has not been able to do so, let alone a 7nm fabrication process. These delays are caused by the fact that intel relies on vertical in-house production of its chips, and it is simply not able to achieve the 7nm level. Intel also spends around ten times as much on research and development compared to its competitors, AMD and Nvidia. By comparison, AMD has managed to surpass intel in various performance checks due to their 7nm processors manufactured with the help of TSMC. Apple has recently cut off ties with intel, releasing its own arguably superior M1 chip. Even Microsoft, a name almost synonymous with intel, have started working on their own custom processors for their devices. What are the consequences of this performance dip? It is worth noting that Intel still holds the largest market share in processors for computing devices. However, it recently had its crown as the U.S's most valuable chipmaker stolen by NVIDIA. The release of new custom processors by companies such as Apple, Microsoft and Google have shown a severe anti-intel movement in the tech industry. Intel has to act fast to get to a 7nm fabrication process somehow, even if that involves partnering with fabrication foundries to produce their chips.

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AI Creativity

Pradhay Amarnath

Unless you're a machine and you're reading this, you are just a human being. You perceive, understand, adapt, and possess the ability to respond to whatever predicament life bestows upon you. Whether you consider yourself creative or not, you develop imaginative solutions to tackle them.

Every situation requires consideration of all angles. Otherwise, we find ourselves in an unruly, chaotic ingenuity whenever a problem is encountered. This methodical problem-solving aspect of the human brain is often affiliated with the term Artificial Intelligence (AI), a type of intelligence creeping into the realm of creativity in Computer Science. The real question lies in whether AI can rival or even surpass human creativity, helping with future advancements. Or is it just destined to expedite and augment it?

To begin formulating the answer, we must first conjecture what creativity is. Creativity is a fundamental characteristic of human intelligence rather than an extraordinary "faculty" or psychological attribute. It manifests in everyday actions like the association of ideas, remembrance, discernment, analogical thinking, and self-criticism. Apart from the cognitive dimension, it's also built around motivation and emotion. Innovation, more often than not, is contextualised by social connections and relationships. When it comes to creativity, we are plausibly safe in our idiosyncrasies, and there are stark differences compared to AI's cognitive approach.

Human creativity descends from a particular inclination towards quandary. Despite lacking social intelligence and interactions beyond the realm of their specific purpose and data set, AI thrives with organisation, patterns, and precision. However, a perfect combination of the two could be paramount for future advancements. AI can help pave the way for enhanced human creativity. In many cases, AI's central provision has been recommendations to human designers, who subsequently settle on the outcome. Hence, it's more apt to challenge us and

guide our creative process; provide us with unique ideas for increased innovation, rather than challenging our prideful hold on creativity. Margaret Boden, a research professor, expressed how she thinks "the ultimate vindication of AI-creativity would be a program that generated novel ideas which initially perplexed or even repelled us, but which was able to persuade us that they were indeed valuable." Therefore, the two primary ways technology can encourage creativity is by saving time for humans to focus on innovation and creatively combining technologies to create distinct working ways.

One instance of creativity showcased by machines is in Go, often acknowledged as the most "challenging classical game for artificial intelligence" due to its sheer complexity. Hence, it was proclaimed inquisitive research at the time. This was until Google-DeepMind introduced the AlphaGo, a program that combined advanced search trees with deep neural networks. It worked by feeding in the board input parameters and was processed through several different network layers containing millions of neuron-like connections. Over time, AlphaGo updated to display better learning and decision-making through reinforcement learning. It could predict expert moves with a formidable accuracy of 57%. Trust me, though this accuracy seems low, it could easily beat amateurs. Why? Well, it was the highest accuracy seen at the time. It could literally "Go" to any global arena and easily defeat world champions. Undoubtedly, it even trounced the world's reigning champion, Mr Lee Sedol, in a legendary game using creative intelligence to devise a winning strategy. Its 37th move characterised the win, a move no human would ever ruminate to make. DeepMind has developed newer versions, including AlphaZero and AlphaGo Zero (also used for games like Shogi and Chess), revealing unconventional strategies and creative new actions, with performances surpassing its former models.

However, this is just the beginning. Companies like MIT are adopting technologies such as natural language processing to create horror stories. Similarly, Google is writing poems with it. Reconsidering how we can draw on AI's creative capabilities to prompt a landscape of creative opportunity is notably essential. Machines are growing better at completing complex, abstract, creative tasks. The implications for the future of work, education, and human societies are profound. Effectuating innovative solutions and pushing creative boundaries is crucial as we navigate the AI era. Thus, it will return the favour by providing the opportunity to develop more efficient solutions for future problems.

AGI -Utopian freefall or Faustian deal?

Amelia Downs

Abstract

Artificial General Intelligence occupies a central space in the field of computational philosophy as the convergence between two key facets of science: computational and cognitive. In essence, understanding the very makeup of the future of AI means grappling with our own sense of identity. It prompts us to ponder academically stimulating questions such as “What constitutes intelligence?”, while forcing us to neatly simplify, reevaluate, and prioritise, our individual and societal needs. Effects on the economy, geopolitics, and perpetuation of bias, for example, will all have to be dealt with when extending our current AI solutions into the omniscient, and subsequently more trusted, AGI alternatives. The further you descend into the black hole of AI, the clearer the ineradicable question becomes: Is this cult of sensationalism which inextricably surrounds coverage of AI unwarranted, or are we somewhat justified in trying to damage control our inevitable freefall into dystopia?

An explanation

When first dipping our toes into the seemingly infinite pool of discussion and debate which is artificial general intelligence, it seems logical to ascertain what AGI means by splitting the phrase into its triad of components.

Artificial simply describes something which is made by humans, often as a copy of something natural. (Cambridge Dictionary, n.d.) It therefore follows that artificial intelligence is the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings (i.e. the product we are producing in this instance is the nebulous “intelligence”, and the copy of something natural is perhaps us, or our own system of thought). (Britannica, n.d.) There is a well-established consensus that AI exists. So, what is AGI?

This integral keyword “general” has been utilised as a differentiator from artificial narrow intelligence (ANI). ANI has arguably catalysed the beginning of the fourth industrial revolution by offering consistently accurate

solutions to narrowly defined and structured tasks, yet AGI promises even more. Artificial General Intelligence is, in short, the ability of a machine to exhibit intelligence on par with that of humans, having, but not limited to, the attributes of common sense, background knowledge, transfer learning, abstraction and causality.

The former two constituents of artificial general intelligence are arguably an issue of semantics; they only have meaning when viewed in respect to what intelligence entails. Hence, we will be dealing with the latter component separately.

What is intelligence – from a psychological and computational perspective

As expressed by prominent psychologist RJ Sternberg “Viewed narrowly, there seem to be almost as many definitions of intelligence as there were experts asked to define it” (Gregory, 1998). This seemingly infinite number of interpretations lends weight to the subjectivity of intelligence, and, subsequently, the difficulty found when trying to explain exactly what artificial general intelligence is.

Definitions of intelligence range from “the ability of an organism to solve new problems” (Bingham, 1937), which discounts the view that computers can have intelligence at all, to “the ability of a system to act appropriately in an uncertain environment, where appropriate action is that which increases the probability of success, and success is the achievement of behavioural subgoals that support the system's ultimate goal.” (Albus, 1991) In general, a computational definition of intelligence seems to prioritise the ability to meet goals in a restricted problem space, whereas a psychological perspective tends to focus on intelligence being adaptation to environmental changes in conjunction with rational thinking. (Ferguson, 2021)

I personally feel that benefit would be found in agreeing upon a universal definition. In ethics generally, the purpose of defining definitions is twofold. Firstly, to make potentially significant distinctions between different types of intelligence, and, secondly, to provide us with a precise lexis so that arguments are less likely to be misconstrued. (Dunn & Hope, 2018) In the context of artificial general intelligence, agreeing on a singular definition will enable us to agree on contingency plans with a strong, collective understanding of cause and effect, as well as allowing experts to make increasingly accurate predictions on when AGI will be realised. We have therefore ascertained that intelligence is grossly overcomplicated.

Even if we did have a distinct definition, determining whether a machine meets the specified criterion would prove difficult. Is there a metric which we can use to quantify intelligence?

The Turing Test

The Turing Test, commonly dubbed the Imitation Game in popular culture, is one such system of measurement. The following game, comprised of a human, an interrogator, and a computer, is detailed. The interrogator inhabits a room distinct to that of the machine and the human. The human and machine are given labels, the former X, and the latter Y. The interrogator states the label of the entity to which they want to talk to, and then asks a question. The aim of the human is to deter the interrogator from believing that they are the human, and the object of the machine is to convince the interrogator that the machine is in fact the human. If the machine passes this test, that is manages to fool the interrogator into believing that it is the human, then the machine can be thought of as intelligent. (Oppy & Dowe, 2021)

Are machines intelligent?

Abiding by the axiom that the machine passes the Turing Test, would it be justified to bestow the honour of intelligence upon it? Literature released by Turing in 1950 mainly concerns objections to the claims that computers are not capable of “thought”, which seems to be drifting into the realms of philosophy concerning consciousness. Despite this, many of these claims can be applied to computational intelligence, and hence will be observed. (Oppy & Dowe, 2021)

The Theological Objection

In the branch of philosophy concerned with mind-body dualism, it is believed that the mind and body should be treated as two separate entities, and such the creation of a body does not automatically guarantee creation of a mind. (Britannica, 2021) It therefore follows that digital computers are indifferent to other material constructs. In terms of theology, the meeting of mind and body could arguably only be achieved by a divine creator. Despite this, there are a multitude of arguments against mind-body dualism, and some against theism. It has even been conjectured that there would be no concrete reason for God to not grant a computer a soul (the supposed epitome of the unification of mind and body). (Oppy & Dowe, 2021)

The “Heads in the Sand” Objection

This is perhaps the most common objection seen towards the intelligence of machines. Proving the ability of machines to think would perhaps engender wide-scale fracture of humanity’s fragile egos. This system of thought is skilfully, and inexplicably, integrated into the core of our identity. It is the crux of our superiority complex, and without it, what are we? Is humanity subsequently futile? And, most importantly, if this is the individual conclusion we come to, is this something we are willing to accept, and what greater impacts will this have on society as a whole? Leading on from this, the development of AGI, or super-intelligent computers, will inevitably excite fear that we may be dominated, or even superseded, by such beings, and this galvanisation may place delicate power structures at risk of implosion. This is a somewhat valid concern, but it is simultaneously an invalid argument. What we have here is a case against the creation of thinking machines, not necessarily their potential existence. (Oppy & Dowe, 2021)

The Mathematical Objection

The mathematical objection towards non-human intelligence is wholly dependent on whether humans are subject to the Lucas-Penrose constraint. To understand this, let’s first ascertain the foundations of Gödel’s Incompleteness Theorem, and thus the Lucas-Penrose constraint in this context.

First incompleteness theorem (with input from both Gödel and Rosser) states that “Any consistent formal system F within which a certain amount of elementary arithmetic can be carried out is incomplete; i.e., there are statements of the language of F which can neither be proved nor disproved in F.” (Raatikainen, 2021) Therefore, an entity subject to the Lucas-Penrose constraint is said to be unable to prove a class of true statements expressible within the system. (Oppy & Dowe, 2021) When viewed in relation to the Turing Test, this implies that for a computer (subject to the Lucas-Penrose constraint) there exists a class of unanswerable questions.

If humans are not bound by this same limitation, then the question which is unanswerable to the computer is answerable to the human. By questioning both entities, then their answer, or, in the case of the computer, inability to answer, will reveal the identity of both, and hence the machine would fail the Turing Test. Despite this, Turing countered that there was no proof thus far that humans weren’t also subject to the Lucas-Penrose constraint, and therefore this objection was deemed invalid.

The AGI avalanche: Dystopia or utopia?

After discounting the previous claims against machine intelligence, we are left to comprehend a future reality where AGI is very much in existence. The words artificial general intelligence seem to be chaperoned by an army of warning signs and that incessant sense of impending doom. However, is AI nothing more than augmented intelligence (as IBM once desperately attempted to rebrand it (IBM, n.d.))?

Dystopia

A popularised thought experiment developed by Nick Bostrom, director of the Future of Humanity Institute at Oxford University, reimagined the future of AI as a herd of paperclip maximisers (Bostrom, 2003); a super-intelligent being that stops at nothing to achieve a goal, in this case the arbitrary creation of paperclips. This highlights a key danger in anthropomorphism. Bostrom prompts us to imagine a realm “in which all possible minds can be represented.

We must imagine all human minds as constituting a small and fairly tight cluster within that space”. (Bostrom, 2012) In essence, even though the fabric of human values seems innate and logical to us, it would be immensely naïve to assume that all other intelligent beings will have the same set of values. Bostrom goes on to assert that “In fact, it would be easier to create an AI with simple goals like (maximising paperclips), than to build one that has a humanlike set of values and dispositions.” Bostrom explains this using the orthogonality thesis: Intelligence and final goals are on orthogonal (statistically independent) axes along which possible agents can freely vary. In principle, except for some caveats, any level of intelligence could be combined with any final goal. In the same paper, Bostrom draws on the instrumental convergence hypothesis which essentially implies that we cannot except a superintelligence with a set goal to exhibit appreciation for human welfare, desires, and survival. An agent with such a final goal would have a convergent instrumental reason to acquire an indefinite amount of physical resources and, if possible, to eradicate potential threats to its goal system and to itself. (Bostrom, 2012) It is this realisation, I feel, which has trailblazed the frenzy of AGI cynicism.

Utopia

Despite this, the emergence of secure ethical infrastructure should hopefully alleviate these concerns. The EU, for instance, has issued a white paper (a method of presenting government policies and legislation for gauging public reaction), which details their approach to future legislature and

regulation regarding AI. (European Commission, 2020). Within this conference proceedings lies an undertone of hope.

The Commission believes that AI has functionality in healthcare (with some already in use for detection of breast cancer (Freeman, et al., 2021)), agricultural efficiency, mitigation of climate change, and security, amongst others. However, the advent of the Singularity, the inflection point of AI gaining consciousness, or an exponential, and potentially irreversible growth in their abilities, will undoubtedly have consequences we need to be prepared for. If we are, by implementing secure AI regulatory bodies in the meantime, we could be facing a future that promises a cure to cancer, world hunger and volatile political systems.

Conclusion

Artificial General Intelligence is perhaps teetering on the horizon of technological advancement, culminating to the cataclysm which will be the Singularity. Or perhaps, as predicted by many experts, it may never be realized. (Fjelland, 2020) As expressed by astronaut Gene Cernan “Curiosity is the essence of the human existence”. Will our conquest to further civilization result in the forfeiture of the integral ingredient to a happy and purposeful life? Artificial general intelligence therefore not only forces us to question what intelligence, and consciousness, is, but it also urges us to reevaluate our purpose. Why are we running? What are we running from? And, most importantly, what are we so desperately running towards: utopia, or just another Faustian deal?

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How does a Machine Learn?

Varun Srinivasan

Artificial Intelligence and Machine Learning

AI and ML are seemingly just buzzwords nowadays, leaving the real essence of this phenomena shrouded and obscure. This article is meant to provide the uninitiated with a brief glimpse into precisely what machine learning is, how machines can learn and what it is capable of.

Types of Machine Learning

Before we delve into how a simple machine is able to recognise patterns and generate predictions, we must establish the 3 main ways that a machine can learn, their aims and how they can be trained:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

This is the most basic type of machine learning. The machine is constantly learning under guidance, similar to how a student may learn under the supervision of a teacher. We use supervised learning to create predictions on large datasets whereby the data is already labelled. During the training phase for this type of machine learning, labelled data is fed to the algorithm and the output is also clearly specified; the algorithm must simply map the input to the output. The primary aim of supervised learning is the draw predictions from a given labelled set of data. Consider this example ... if we have a dataset labelled with:

- Height
- Weight
- BMI
- Sex
- Cardiovascular Disease?

If we want to predict whether an unknown test subject has or does not have CVD, given the other factors, we can feed a trained supervised ML model (such as a Random Forest Regressor) to make this prediction.

Unsupervised Learning

Unsupervised learning is a clear step up from supervised machine learning. In this case, the machine is fed unlabelled data – it is not told what the output or input looks like yet will still have to find patterns in the data in order to create predictions on the test data. The training phase for unsupervised learning is much longer than for supervised models as the machine is only given the input and will have to find the output on its own. Going back to example above, in an unsupervised environment, the machine will be fed a photo of a person and will have to recognise patterns such as:

- Height
- Width
- Sex
- Perceived weight

Based on this information it will put the person into a cluster (using a K-mean cluster model) to indicate whether the person has CVD or not.

Reinforcement Learning

So far, we have only discussed the use of Machine Learning in fixed data sets to predict quantities such as BMI. However, reinforcement learning completely redefines the capabilities of Artificial Intelligence: in this type of learning, machines are able to interact with the real environment around them and produce almost super-human feats. The goal of RL is much different from the goals of the other 2 machine learning types.

Apple example:



Apple example:



Apple example:



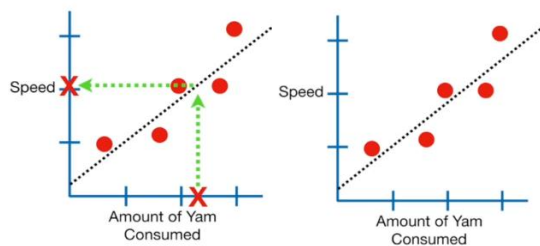
In Supervised Learning, the model is fed a set of inputs (x) and it needs to produce a binary output (y) which indicates whether the thing identified is an apple or not

In Unsupervised Learning, the model is fed a list of inputs (x) and it needs to find patterns between the 2 images. It does not know that they are apples but identifies that they are similar

In Reinforcement Learning the objectives are different and therefore the output also differs greatly. The aim of RL is to maximise rewards, which in this case would be to survive. Hence it needs to identify what objects in its environments can be used to achieve this aim, and the actions that have to be taken to do it, which would be to eat the apple.

A gentle introduction

As explored above, the main aim of machine learning is to train a model using training data in order to predict a value given testing data. There are several new, complicated, and confusing machine learning models such as the Deep Learning Convolutional Neural Network, but in order to give a simple, yet insightful foundation on the subject, we will use a regression model to find the link between how much yam someone consumes and their speed. This is merely fitting a line of best fit to a given set of data points.

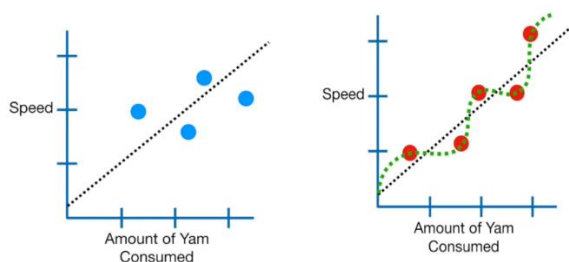


From the red dots on the left – **training data** – it is evident that there is positive correlation between these 2 factors, as further emphasised through the black line of best fit.

However, that line is not just to show the relationship, but is primarily used to predict values i.e., the speed, given the amount of yam consumed.

However, a line of best fit is not the only way to fit the training data; instead, a line connecting each of the training data points could also be used ... surely that would give more accurate predictions as it is closer to the training data?

The green line is said to have a high bias as it fits the training data almost perfectly, whilst the black line has a low bias due to the larger difference between the fit and the actual values.



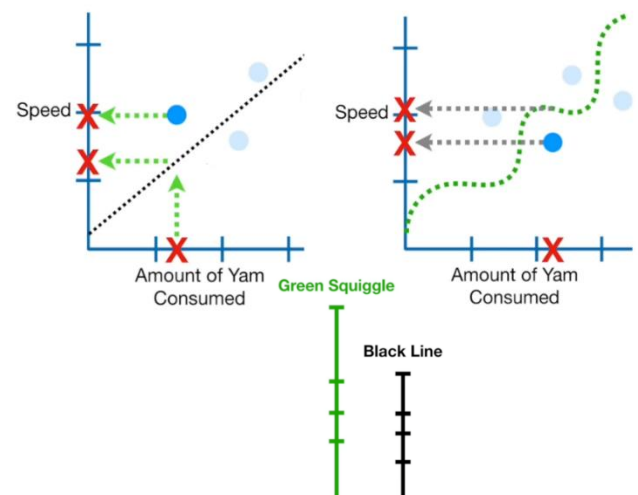
The only way to assess which line is better – the straight one or the green squiggle – is to **test** the model. During the testing phase, unseen data - **test data** - is fed to the machine in order to generate predictions, as represented as blue dots on the right.

The machine will then use a **loss function** to find the **variance** between the predicted value for speed and the actual value. The greater the variance the greater the inaccuracy of the model

In this case we will use a loss function of $(\text{actual value} - \text{predicted value})^2$ as it ensures that negative values do not cancel out the positive ones.

The diagram on the left allows us to visualise the difference between the predicted value (indicated by the black line for that X value), and the actual value (indicated by the blue dot). This process is repeated for all of the values in the test data to find the total variance for the black line. Following this, the process outlined above is repeated on the green line to find the total variance for the green line.

The diagram on the left is a visualisation of the variance for each of the lines. It can be concluded that the total variance is much greater for the green squiggle compared to the black line; thus, it is a worse model for making predictions. Machine Learning is all about making predictions; although the green squiggle is better fit to the training data it performs worse on the testing data, so it is not the desired model



Cross Validation, Bias and Variance

Cross-Validation

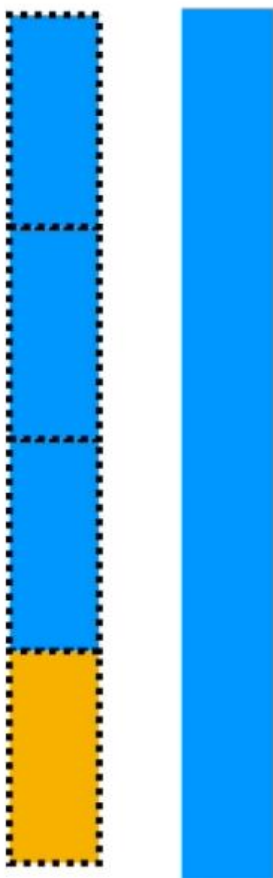
Let's say that we have a dataset specifying people's:

- Height
- Weight
- Waist to Hip ratio
- Sex
- Whether they have CVD or not

If we wanted to predict whether a person has CVD based on the other 4 factors, there are several machine learning methods that we can use (K-mean clustering, Logistic Regression, Decision Trees etc), but how do we evaluate the best method, i.e., the method which produces the most accurate predictions?

This process is called **Cross Validation**.

Cross Validation allows us to compare different machine learning models in practice to determine the most effective one to be used in a real-life scenario.



If the rectangle on the right represents all of the data that we have on the parameters stated above, regardless of the model that we are using, it must be first:

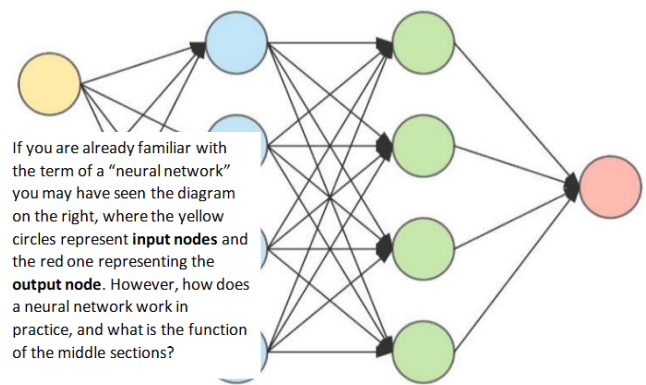
- Trained using Training Data
- Tested using Test Data

Data must be split so that some data is used for training, and some is used for testing, such as a 3:1 split. 75% of the data is used for training the model and the rest is used for testing as seen on the right.

Similarly, how do we know which block of test data is the optimum way to divide the data? Some models may perform better on the top 25% of the data instead of the bottom 25% and vice versa. To resolve this issue, Cross validation uses all of the sections, one at a time for training and testing.

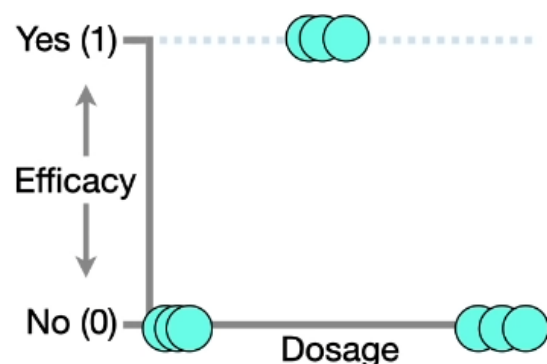
The error of predictions is calculated for each block for each model and the total error is calculated when all of the blocks have been for testing for each method. Taking the example above, the total error for K-mean clustering is compared with that of Logistic Regression and Decision Trees. The method with the lowest error is taken forward as the primary model to use for further predictions.

Neural Networks – A brief intro



Once again, let's delve into this topic with an example. A vaccine has been developed to combat the outbreak of a new disease, and scientists need to find the optimum dosage. They have split volunteers in 3 groups:

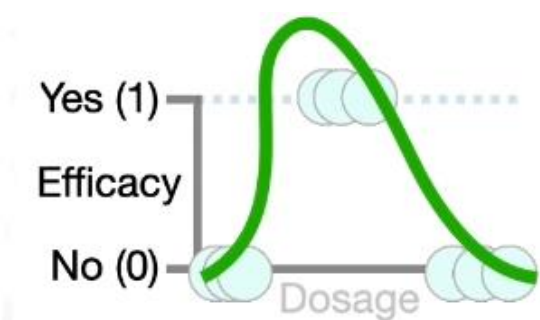
- Low dosage
- Medium dosage
- High dosage



From the data collected, they discovered that low dosages were not effective (which we can set to 0 efficacy); the medium dosages were very effective (1 efficacy) and finally the high dosages were not effective (0 efficacy). A graphical form of the data would appear similar to this.

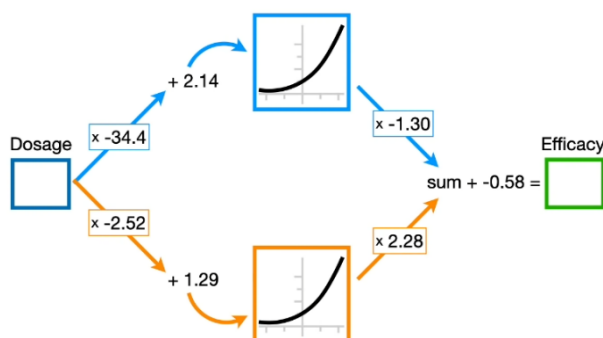
However, the main essence of ML lies within creating predictions. Given a dosage, what is its efficacy? To answer this question, a graph has to be drawn to fit this data, which is exactly what neural networks can do. Although this problem is relatively simple due to there only being 1 input parameter, in a face recognition system, neural networks are paramount in tackling the inherent complexity of the inputs and the output.

Whilst most machine learning models are only able to fit straight lines to the data, neural networks are able to fit squiggles to the dataset, which is required in this scenario.



To explore how this neural network works, we will be using the diagram on the left. A neural network consists of nodes (the Dosage, Function Graphs and Efficacy), and connections between the nodes.

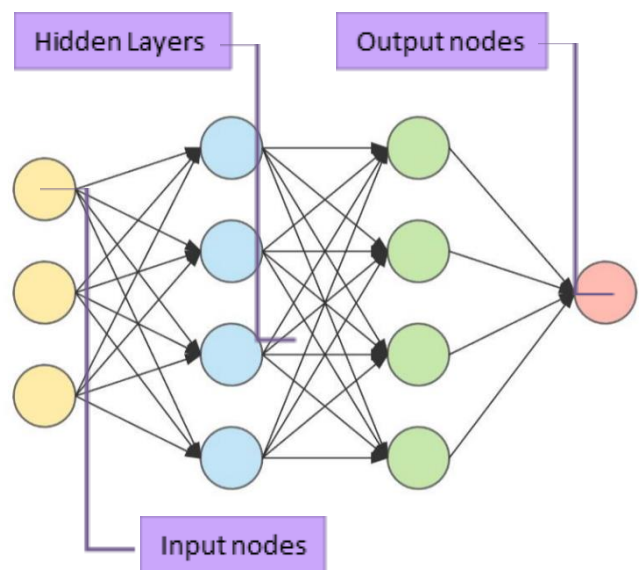
The connections store the parameter values which are analogous to the **gradient** and **y intercept** of the graph. Initially these values were estimates, which were fit to the data through a process called **backpropagation**. For this example, let us assume that the parameter values have already been fit to the data using this process.



The curved lines in the middle nodes are called activation function; they enable the neural network to fit a curve to the original dataset by finding the values of the curve. There are 3 main activation functions used in practice:

- ReLU
- Sigmoid
- Soft plus

Each one has a different shaped curve and hence equation, but they all serve the purpose above. In this case we will be using the soft plus activation function. Here there is only one input node and one output node, but in practice, neural networks will often consist of many input nodes and several layers in between the input and output nodes. The layers in between are known as the hidden layers.



How does a Neural Network Learn?

Going back to the example, if we label low dosage as 0 and high dosage as 1, we need to create a model that can predict the efficacy of any dosage in between. But how does a neural network do this?

Initially the neural network follows the blue connection, inputting values in increments of 0.1 between 0 and 1.

It uses the formula $Xval = (-34.4 \times Dosage) + 2.14$ to find the X value on the Soft Plus activation function, which is $f(x) = \log(1 + e^x)$. Each y-coordinate value is then multiplied by -1.3 in a process called scaling to complete the first transformation of the soft plus curve.

After completing the above step, we end up with this light blue curve, and now we repeat for the orange connection.

For the same input Dosage values, the

$X_{val} = (-2.52 \times \text{Dosage}) + 1.29$ and the corresponding y-coordinate value is found using the soft plus function $[y = \log(1 + e^x)]$

Finally, the y values are **scaled** by 2.28 to produce the orange curve below.

Summary of Neural Nets

Neural Networks start with a basic activation function such as soft plus or ReLU, however, the weights and biases calculated through **backpropagation** are able to stretch and transform the curve to fit the data provided.

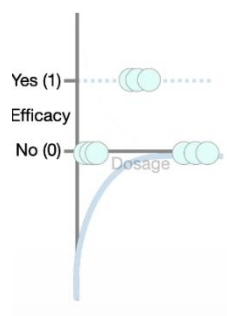
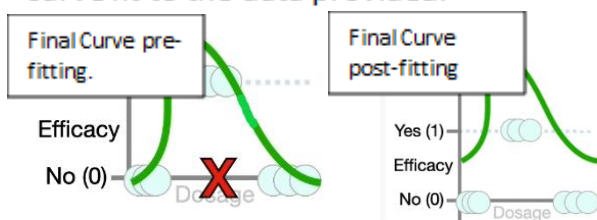
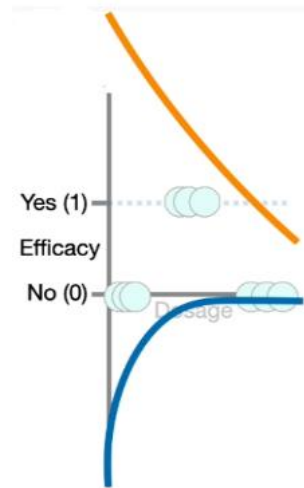
The new curve is then able to create predictions on a dependent variable given any independent variable.

That is all there is to a neural network!

Finally, the neural network adds the y-coordinate values from the orange curve to the blue curve and this curve is once again scaled by a given value to give a curve fit to the data provided.

Conclusion

Machine Learning is used everywhere around us in our day to day lives, from recommendation algorithms to self-driving cars and fraud detection, yet still remains a mystery to many people. This article should have given an introductory read on the inner workings of Linear Regression and Neural Networks, the 2 most important machine learning models.



Enhancing Musical Experiences with Technology

Joshua Senoron

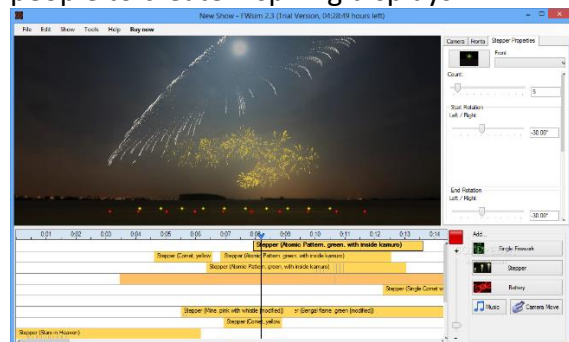
Here is something that's quite fascinating with seeing such spectacular displays sync up to music. For example, millions around the world watch in awe at midnight to watch the New Year's Fireworks shoot up into the night sky.

I mean, there's something so magical in seeing spectacular bursts of light in the sky as we hear music swell in intensity, inspiring us all to prepare for the year ahead.

Where technology enhances music experience physically

Computers play a key role in these events, making it possible for technical teams to preprogram the songs and firework effects in advance. With the help of technology, stunning displays can leave us all with long lasting impressions.

Software allows designers to simulate how the fireworks will look, allowing them to experiment with ways that they could evoke the most nostalgia for audiences, by syncing beautiful displays to beloved songs. The best part? It's that no one in the creative team needs to worry about the physics and mechanics of it all, and so, technology is vital in abstracting away the unnecessary details which may hinder the creative process, allowing incredibly talented people to create inspiring displays.



Demonstration of FWsim, which allows anyone, not only pyrotechnicians, to simulate fireworks if they so wished

As well as this, engineers can also utilise music technology to brighten up the everyday. In 2009, Volkswagen set up an experiment which reimagined a staircase as a piano, where each step had a pressure sensor, playing the corresponding note when activated. This idea was put in place to test their hypothesis that if healthy-yet-difficult activities (taking the stairs) were made more fun, people would favour it over easy-yet-mundane methods (the escalators). And it worked!

It's a small gesture, but the use of music and technology in this way made a huge impact to the everyday decisions of commuters, promoting a more active lifestyle and overall better wellbeing.



These stairs are installed at Odenplan metro station in Stockholm, Sweden

How technology enhances music creation

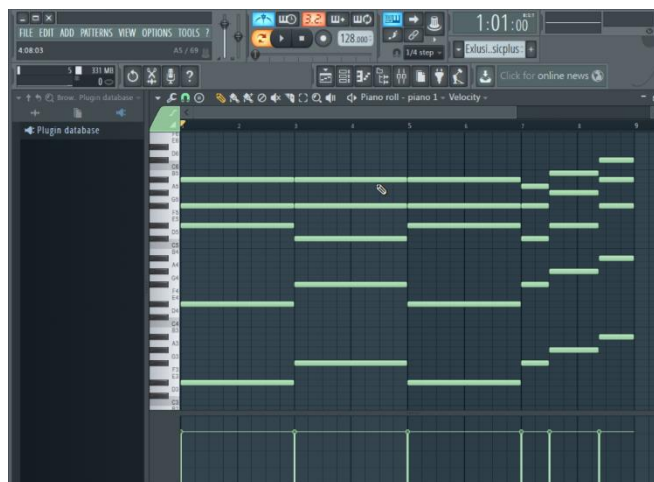
It's exciting to see how music can be presented in the real physical world, but the possibilities open vastly for musical experiences in the digital world. For example, MIDI, (Musical Instrument Digital Interface) which is a standard protocol from the 1980s, allows for music to be stored in a format which virtually any electronic instrument can understand.

Essentially, MIDI files contain timestamps for states when a note should be "on" or "off". An "on" state essentially indicates that a note should be playing. The files also contain information about volume, instruments and even pitch bending, making MIDI a super simple protocol which is advanced enough to make it possible to

transcribe most songs into a digital format. With practically every audio editing program allowing MIDI inputs, composers and performers, for the past 40 or so years, can pass around musical data around different programs with ease.

For example, recordings from electric pianos, can now be purely stored as a sequence of notes, instead of complicated waveforms, which may contain impurities such as background noise. Taking the exported MIDI data, I could then import it into any musical program of choice and edit it in any way I please. Through MIDI, it's a trivial task to digitally edit recorded passages of music, by editing pitch, rhythm, or even switching the instrument entirely.

If I wanted to, I could plug in a custom synthesizer to playback my notes, creating new and interesting results. As you can see, MIDI proves to be an incredibly useful tool, for modern music creators everywhere.



In FL Studio, you can record chords directly into the program by using a MIDI controller, such as an electric keyboard.

Where technology enhances music appreciation

However, not only is MIDI useful for those involved with music production, but it also has great strengths in helping listeners understand the music they listen to. As shown in the previous image, the “piano roll” style of MIDI visualisation is a very popular and intuitive way of showing what kind of notes get played on screen; high notes on the top, low notes on the bottom.

However, a guy called Stephen Malinowski, was one of the first to represent music in this style, being an early pioneer in reimagining a music score in a visual digital format.

Similarly, as seen in the image previously, Malinowski’s works involved condensing daunting and note-dense sheet music, into modern, simple to follow animations.

Much like a traditional music score, these animations display pitch along the y-axis and time along the x-axis, but with extra changes. In a traditional music score, if someone wanted to follow along with the score of a complex symphony, for example, they would be required to mentally combine music from multiple lines at once, keep track of which part played which and pray that they didn’t lose their place. It’s a manageable task for musicians, but considerably difficult for most people, and this difficulty could often put them off from Classical music, dismissing it as boring.

However, Malinowski solves this by reducing the complexity of a score to only a single line, where colours are used to tell different instruments apart. There is also the added creative freedom of being able to experiment with colours and shapes to explore how melodies can interact with one another on the score.



Here is of comparison between the two visualisations from the first movement of Beethoven's 5th Symphony

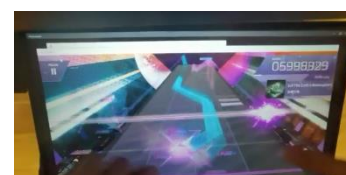
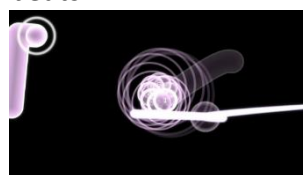


Although these animations may miss out some of the nuance from a traditional score (such as dynamics), are such details really necessary for an audience member? Even for those without musical training, the prominence of the famous *short short short long* motif is immediately obvious! Anyone can see that the prominent repetitions and slight variations of this motif explain why Beethoven's 5th Symphony is such a coherent and famous piece (and even if you don't know the name, the powerful melody is definitely something you'll recognise!). Over the years, many people have expressed their thanks, finding that these animations have changed their experience of Classical music. The nature of having auditory feedback confirmed by the flashing notes on screen in such a satisfying fashion, do wonders to help anybody intuitively follow along with the score, helping them appreciate the music in a way they otherwise wouldn't have. Due to their simplicity, these animations have been used as a teaching tool for children, providing their first experiences of relating sound to a spatial structure. In fact, Malinowski himself is a co-founder for an association called *Music:Eyes*, which aims to provide music education for children, helped through the use of his fun animations. Personally, I find these animations very engaging, and I sometimes I find it fun to plug in a few MIDI files into Malinowski's program to see the results on display! With such a positive reception worldwide, it's no wonder his YouTube channel (smalin) has amassed over 170 million views.

Other Notable examples

The examples I've showcased here are only a small fraction of how technology has really impacted and changed how we listen to music. I didn't even include the impact of streaming services such as Spotify, which has made listening to music a widely accessible pastime. Long gone

are the days of purchasing physical CDs, when nowadays your favourite songs are a tap away. In fact, 2020 saw Spotify reaching more than 300 million active monthly listeners! As well as this, the development of music technology has even grown into the gaming industry to create a fusion of genres, and a brand new kind of game: rhythm games. Remember about the example where I mentioned how engaging it was for someone to follow along to music if they had visual feedback synced up to the beat? At this point, it is no surprise that games which exploit this mechanic find international success, such as *Dance Dance Revolution* which sold 6.5 million units worldwide. I'm pretty sure that most of you have even come across one of these machines at arcades around the country! As a matter of fact, the rhythm game industry still maintains its high relevancy over the past decade, with later titles such as *Arcaea*, *Osu* and *Geometry Dash* engaging the masses with addictive gameplay and banging beats.



Gameplay courtesy of Jaydn Li and Harry Chen

Conclusion

In conclusion, it's clear to see how much technology has impacted everyone's experience of music, whether it be through physical or digital means, how it's changed the way we listen or create, and how we learn and play. Even if you aren't a huge fan of music, I'm sure that you've noticed the incredible rise of technology, and how it's influenced our sonar world. How will it change in the future? How else could music influence us? Could the development of AI play a significant role in music creation? I don't know, but it all **sounds** pretty interesting to me.

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Where to download the Malinowski's MIDI player
<https://www.musanim.com/Player/>

Video Games – The key to our future?

Shravan Bharadwaj

I like video games. I'm not good at them, but I am good at procrastinating, so I end up playing them a lot. After sinking countless hours into 'maths homework' and 'IPM revision', I wanted to be better... not at finding the area under curves or writing more efficient code, but at planning areas in MegaMansion #14, my latest and greatest Minecraft creation capable of housing the entire Indian subcontinent and increasing efficiency in my Tetris play even though the doctor says if I get any more carpal tunnel, he'll have to amputate my arm (it's a small price to pay). Eventually, this obsession with improving had me stumble across less traditional methods: solving games. A solved game is "a game whose outcome can be correctly predicted from any position, assuming that both players play perfectly." I had found the key to my future; I could win. All I had to do was ~~input computationally perfect moves in a manner completely devoid of fun~~ be the better player and win through lack of counterplay for those unlucky enough to oppose me sheer talent and hard work.

Why stop at video games?

If this could be applied to life: the greatest role-playing game that will ever exist, achieving utopia would be as easy as sequentially carrying out pre-determined actions. A brighter future for everyone. Where would I even start? How could I even begin to tightly represent the fluffy concept of social interaction? I was stuck trying to reinvent the wheel when I realised much of the groundwork had already been done for me. John von Neumann, and his book 'Theory of Games and Economic Behaviour' (1944) coauthored by Oskar Morgenstern, founded the field of Game Theory as a mathematical discipline and proved his minimax theorem.

The book established that in zero-sum games with perfect information, there exists a pair of strategies for both players that allows each to minimize their maximum losses. Simply, this means in a game one person's advantage is another's loss (zero-sum) and no information is hidden from either player (perfect information), both players have an equal chance of winning.

The minimax theorem, despite its use of confusing words, is flawed: it has very little practical application. In the real world, very few interactions are contained between two entities and even fewer have perfect information. An accurate abstraction of a real-world game would be complex non-cooperative interactions between multi-agent systems where each player's strategy changes depending on that of others, which Neumann's theorem simply couldn't begin to solve.

Luckily, amidst that even more confusing string of long words emerges a solution: the Nash equilibrium (1950) tackles just that. The constraints for the Nash equilibrium to work are remarkably true-to-life, where a player whose strategy is working cannot change (why would any rational player do that?) and, after looking at all other players' strategies, each player cannot find a more effective strategy for themselves (if a better strategy exists, why not use it?)

With social interaction abstracted into tighter models thanks to the efforts of "The Most Original One" (John Nash's nickname at Princeton – not sure that's a compliment) only one thing stands between us and utopia: learning how to actually solve games. To do this, there are some definitions you need to know first. A game tree is a directed graph where each node is a game state, and each edge is a "move" or action the player can take within the fixed constraints of the game. There are 3 ways to solve two-player games: ultra-weak, weak, and strong. Ultra-weak solutions simply proves whether the first player will win, lose, or draw from the starting position, given perfect play on both sides; a list of moves need not be provided. As the more astute amongst the readers may have inferred from the name, this type of proof isn't particularly useful for more complex games, as knowing a possible outcome, but not knowing which of the e.g. 5×10^{20} moves for checkers to play (where each game has on average 25 moves per player) is 300,000,000,000,000,000,000,000,000 (3×10^{26}) times less useful than telling someone they're going to win the EuroMillions jackpot without telling them the winning numbers, and them guessing the winning numbers first try. This is where more complete forms of solutions come into play. A weak solution produces an algorithm to ensure victory for one player, or a draw for either, given perfect play from both players. This is far more useful than ultra-weak solutions as the algorithm provides the player a move to play for every game state. You may be thinking, this is something a 5-year-old could do, and you'd be right: it doesn't take a genius to play any random move per turn, I can think of more than a few babies capable of

losing without breaking a sweat (and a few dogs actually), which is why for a weak solution to be formally accepted, not only does the move have to be the best for the game state, but it must be accompanied with proof proving why each move is the most optimal for the player making it. This raises a potential issue of having to prove an enormous number of optimal moves, which is why the produced algorithm can only be applied to games with perfect play from both sides. Another problem. This is oddly like when I'm coding. Top tip: each fixed bug is secretly 10 more down the line so never fix bugs, implement new features. Even better yet, never write buggy code. As much as it pains me to say this, my boundless wisdom may not be applicable at this next level: strong solutions, an algorithm that can produce perfect moves from any position, even if mistakes have already been made on one or both sides. This could be a difficult problem to solve right?

Wrong.

It's almost trivial for an algorithm to exhaustively search a game tree and find a perfect move – the “brute force” method. The real issues start arising when time becomes a factor. For many non-trivial games, such an algorithm would require an absurd amount of time to generate a move in realistic timeframes, not to mention the huge pre-generated database of all possible game states taking up massive amounts of space. For example, such a database for checkers contains 443,748,401,247 positions. Unfazed by the gargantuan task large enough to scare away the rest of the world, Jonathan Schaeffer and his team conquered the monster that is checkers in 2007. They spent 18 years running on average 150 computers simultaneously to carry out the necessary computations to produce the final algorithm. And what is the result of this algorithm you ask?

A draw.

To date, checkers is the most complicated game solved, but many more are “partially solved”, meaning the game is solved for some, but not all, positions. To solve life, this approach may be the most palatable, since there are more iterations of chess than atoms in the observable universe, (known as the Shannon Number, estimated between 10^{111} and 10^{123} with only 10^{81} atoms making up the observable universe) life is infinitely more complex than a simple game of chess. Therefore, instead of blindly traversing through every node in a game tree in a serial (linear) fashion, the alpha-beta algorithm (John McCarthy 1958) reduces this number by keeping track of how favourable each move is for both players as it guesses future game

states, and if the algorithm detects its opponent ends up better off after a certain move, it prunes that entire branch of the search tree to increase efficiency during runtime. This very algorithm was used in the historic 1997 chess game between IBM's supercomputer “Deep Blue”, running a parallelised version of the alpha-beta algorithm (allowing it to search through the tree hundreds of times faster than the original algorithm proposed), and the then World Champion, Gary Kasparov. This game was the first time a computer had beaten a reigning chess champion under standard time constraints and, despite Kasparov famously saying, “I hesitate to call Deep Blue intelligent... it is as intelligent as our alarm clock” after his defeat, this was a marked shift in the computational abilities of artificial intelligence and since, partially due to the media attention Deep Blue's victory brought, improvements in this field are rapid.

Currently, the best-known game solving algorithm is the Monte Carlo Tree search algorithm, which uses batching (it runs without user intervention as resources permit) to randomly sample edges around the node it's currently on and then calculates how good the moves are by running evaluative functions. This process is far superior to serial processes like the minimax algorithm due to not only cutting down the total number of game states in chess from between 10^{111} and 10^{123} to only 10^{40} “realistic” game states, (a 10^{83} improvement) but also sharing the alpha-beta algorithm's ability to be parallelised, while, unlike the alpha-beta algorithm, being able to adapt to any game state through its real-time random sampling of nearby nodes.

This algorithm is used by Google's AlphaZero, the best chess playing entity in the history of the game, and whilst not solving chess by the definition, it emulates it with almost eerie success. It was given nothing but the rules of chess, and in 2 hours was able to beat every other chess playing engine (program), even those with more than a decade of game data and optimisation time. The approach AlphaZero uses to tackle this unimaginable large problem is ingenious, and whilst solving life may have initially just been a hollow dream thought up by a chronic procrastinator with way too much time on his hands, this gives us real hope.

The future of solving games won't be easy, but it's the key to the future of humanity. Maybe you could be the one to revolutionise the field, maybe I could be the one to change the world. It sure sounds like fun – I'll be sure to get started on that after just one more game.

A brief history of C++

Rahul Raghavan

C++ is one of the most popular languages in this day and age, alongside python and java being some of the other ones. But the reason I want to talk about C++ specifically is that it has an incredibly interesting history, being created in 1998 by Danish Computer Scientist Bjarne Stroustrup as an extension of the C language; he wanted an efficient and flexible language like C that also provided high-level features for program organisation. While the C syntax was still able to work with this language, it brought along many quality-of-life improvements, some of which including vectors, STL arrays and more. He did this during his thesis.

Advantages of C++ over C include being a highly portable language and is often the language of choice for multi-platform app development. C++ is an object-oriented programming language and includes classes, inheritance, polymorphism, data abstraction and encapsulation. This made it far more versatile to work with compared to C and made certain programs become much simpler to write about. C++ has a rich function library, allows exception handling, and function overloading which are not possible in C. In addition to this, C++ is a powerful, efficient and fast language. It finds a wide range of applications – from GUI applications to 3D graphics for games to real-time mathematical simulations, making it the preferred language for designing games along with C Sharp, as well as embedded systems.

In 1998, the C++ standards committee published the first international standard for C++. The Annotated C++ Reference Manual was said to be a large influence in the development of the standard. The Standard Template Library, which began its conceptual development in 1979, was also included. This popularised STL arrays, which is a combination of regular C-style arrays and vectors. In 2003, the committee responded to multiple problems that were reported with their 1998 standard and revised it accordingly.

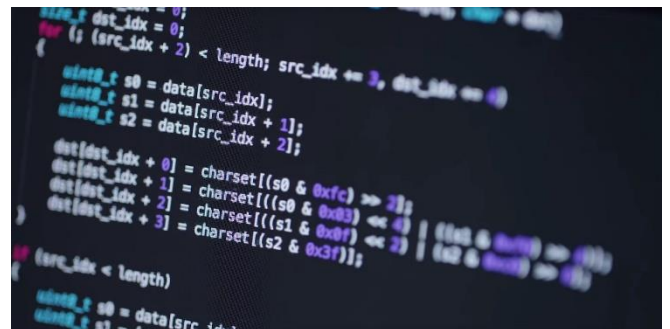
In 2005, the C++ standards committee released a

technical report detailing various features they were planning to add to the latest C++ standard. The new standard was informally dubbed C++0x as it was expected to be released sometime before the end of the first decade. Ironically, however, the new standard would not be released until mid-2011. Several technical reports were released up until then, and some compilers began adding experimental support for the new features.

In mid-2011, the new C++ standard (dubbed C++11) was finished. The Boost library project made a considerable impact on the new standard.

Some of the new features included regular expression support (details on regular expressions may be found [here](#)), a comprehensive randomization library, a new C++ time library, atomics support, a standard threading library (which up until 2011 both C and C++ were lacking), a new for loop syntax providing functionality similar to for each loop in certain other languages, the auto keyword, which could be assigned to any type of variable, new container classes, better support for unions and array-initialisation lists, and variadic templates.

C++ is still growing and getting updated, to the point where specialist programmers don't even consider it a subset of C anymore, which clearly shows how much this language has evolved.



Ethical Hacking

David Wu

You may have come across the term “Hacking”. Commonly known, it is the gaining of unauthorized access to data in a system or computer. The consequences are limitless, ranging from gaining your credentials to bank accounts, steal money or make purchases, to selling your information to other parties for illegal purposes. To combat this crime, everyone across the globe takes many precautions, whether it would be using antivirus software or... Utilizing Ethical Hackers. What are Ethical Hackers?

Ethical hackers are people who have been allowed to gain unauthorized access to data in a system or computer. This means that they can hack on purpose with permission in order to reveal vulnerabilities to the system owners. The process involves finding and exploiting such weaknesses to determine if the unauthorized access is possible. As a result, ethical hackers can help prepare large companies for a cyber-attack and demonstrate the methods used by cybercriminals.

Ethical Hacking Techniques

The goal of the ethical hacker is to mime a genuine attacker. In doing so, they look for attack vectors against the target. The goal is to research, gathering as much information as possible. When the ethical hacker gathers information, they attempt to exploit that to discover vulnerabilities. This could be an assessment including automated and manual testing.

Next, the ethical hacker uses exploits against the said vulnerabilities to prove how the malicious attacker could do the same. For example, the ethical hacker could try injection attacks (Attacker introducing code into a

vulnerable computer program, changing the course of execution).

Finally, once testing has been completed, the ethical hacker writes a detailed report which commonly includes steps to compromise the vulnerabilities, steps to patch them, and the vulnerabilities themselves (I.e., Broken Authentication, or Sensitive Data Exposure)

Advantages of Ethical Hacking

1. Discovering Vulnerabilities from an attacker's POV so the weak points can be fixed.
2. Implements a secure network that prevents security breaches
3. National security from data terrorists 4: Real-world assessments

Disadvantages of Ethical Hacking

1. Limited Scope (Ethical hackers cannot progress beyond a defined scope to make an attack successful)
2. Resource Constraints (Malicious hackers don't have the time constraints that ethical hackers have. There is also computing power and budget)
3. Restricted Methods (Some organisations ask experts to avoid test cases that leads to the servers crashing like Denial-of-service attacks which shuts down machines/networks)



Computer Programming in Sport

Henry Etherington

Computers are used in an amalgam of ways in sports today, and I certainly cannot address every miniscule use in individual lesser-known sports. I will discuss three areas that pan across many sports as well as some other niche uses in the final section.

Sports require training to improve, and a human coaches can only be so good. With sport-specific robots, training can be enhanced beyond human capability. They can be designed in many ways to do many things.

Most commonly, they are used to feed an object such as a baseball, shuttlecock, or ping-pong ball. For example, the respected table tennis robot Amicus from Butterfly is a ping-pong ball feeding machine. It has an easy-to-use interface on a special app downloadable from many app stores usable to program the robot to feed however you could think of for a ping-pong ball, including spins and speeds that couldn't be generated by a human. Furthermore, they could be used to mimic the actions of a simultaneous player of the sport, such as a badminton opponent based on averages from analyses (see below).

Related to training, analysis of sport is vital for improvement. It can precisely show strengths and weaknesses of a performance, just through a simple algorithm of data collection and mapping. It would, however, be hard for a human to keep up live while performing all the calculations to fully analyse many sports. Inputting simple notes into an efficient interface (or even having a computer 'watch' the sport) and computerising all the calculations faster than any human is a beneficial alternative that isn't game breaking.

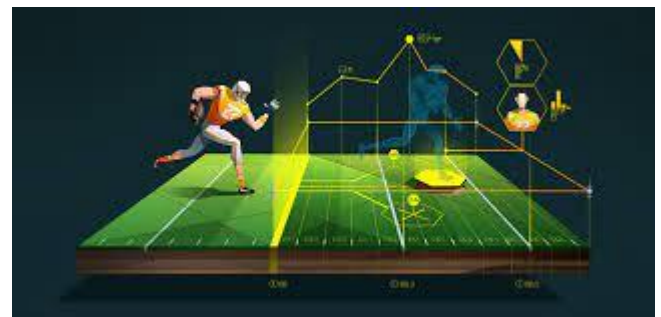
Additionally, it isn't difficult in the least to program the necessary algorithm. I ran Python and challenged myself to program analysis for cricket, tennis, and football as fast as I could; and I managed full analyses for all three with full interfaces in just under an hour.

Humans don't come near the best individual senses of any animal, but a mix of them all. But how do we compare to computers? It turns out, similarly as to the natural world. Almost all sports contain a border or outline – or just some line where we need to see how an object positions in comparison to it. Take LBWs in cricket, where before computers it was down to referees to decide whether the path of the cricket ball beyond contact with the batsman would continue into the stumps.

With Hawkeye technology, we can now picture the path of the ball digitally and receive a near perfect image of the path of the ball. The same goes for line calling in many sports such as racket sports and pitch-based ball games. There are also microphone systems such as ultra-edge in cricket and let calling in tennis that listen for sounds unintelligible to us.

Computers are sometimes used to test equipment (maybe even something else computer-based...) by supplying theoretically possible inputs and/or conditions that test the limits of said equipment. They can also be used to create models of sport where spacetime can be controlled and hence model new strategies and tactics. This is slightly going off on a tangent, but some sports such as rugby and football often involve injury that can cause a player to go out of action.

Modern technology can be used to limit these waits. Onto robotic sports, many robotics enthusiasts compete in programming, computer aided design, and controlling custom robots in school-age competitions such as VEX robotics. Additionally, we have the increasingly popular BattleBots and its predecessor Robot Wars featuring legal bot fighting for entertainment and prize money.



The anatomy of a modern smartphone

Eashan Rautaray

Our lives revolve around technology these days, and at the centre of this crazy technological storm lies just one central figure – a pocket-sized device that's constantly evolving, constantly changing and pretty hard to live without, that's right, it's a smartphone – but what's actually inside these mini-powerhouses?



A chipset: This piece of essential equipment is inside every phone. From Apple's latest A15 bionic chip and Qualcomm's powerhouse, the Snapdragon 888 to Samsung's Exynos series, these chips all contain a CPU (Central processing unit) with often up to 4 cores and 3ghz of processing power, specs comparable to a mid tier PC processor, a GPU (Graphics processing unit), often with the specs, such as the clock speed and CUDA cores being pushed to the absolute limit in such a tiny chip space, primary memory, such as the RAM – often up to 16 Gigabytes of storage, ROM, and a modem, allowing the phone access to the internet connectivity.

A battery: a battery takes up the bulk of the phone. Simply put, a battery works by transferring lithium ions from a negative store to a positive store through a transfer while a phone is working and while charging the reverse occurs, as the charger transfers ions from a positive store to the negative. Battery power is measured in milliamp hours and is essentially how much 'energy' a battery can store. A high-end flagship today might have up to 5000 mAh battery.



Sensors: Phones require multiple sensors to perform multiple everyday tasks. For example a gyroscope is needed to track the phones rotations, an accelerometer

is required to track the phone's movement, biometric sensors that are used for finger/ facial recognitions and magnetometers which measures magnetic fields and compass directions. On top of this a phone may contain smart sensors, which allow AI to measure a room, or a light/ proximity sensor which can track where a phone is in relation to your body.

Cameras: these contain a few main parts: lenses, which controls the field of view (the wideness of an image), aperture, which controls the



brightness, a sensor, which converts the photograph into digital signals. On top of this cameras have certain features which work in combination with the sensors. For example, infrared sensors could be used to detect humans and this tells the camera to increase/ decrease the brightness of the image.

Display: One of the most important features of a smartphone is the display. It contains three layers: a protective outer layer, an interactive layer which registers touches from fingers and a screen which actually displays the images. The display uses LED backlights to illuminate the screen (More often OLED, also known as Organic light emitting diode). Specs of a display to look out for can include: refresh rates (how many times a second the display is refreshed), resolution (how many pixels are displayed in an inch), brightness, colour contrast and response time.

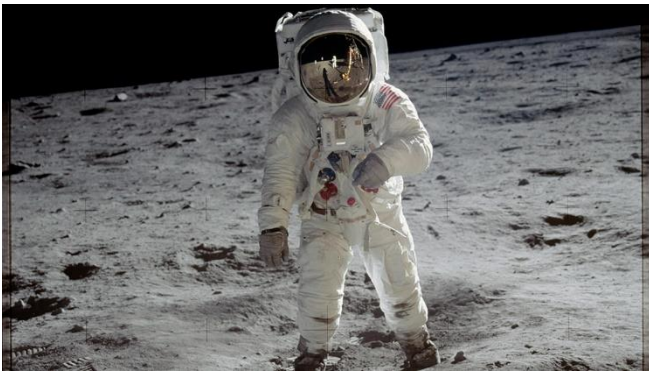
Speakers and ports: phones mainly have one charging port and a headphone jack hole. Often phones come with features that can include ports such as different ports than enable technological ecosystems to work together (phones, styluses and wireless earphones etc.). Front facing speakers work by converting electrical signals into sound. Headphone jacks work a similar way, except the signal is only activated when a pin is plugged in.

These are the basic components of your pocket machine; I hope you enjoyed delving into the depths of your smartphone.



AI on the Moon

Alex Mo



Space is a hellish, vast, empty wasteland, a far cry to the warm and habitable Earth. It seems daunting, and it is very much so, but why are we looking up to seek a future among the stars?

Why we bother

With places to expand on Earth running out, the only reasonable decision is to colonise other space bodies. This also comes with a variety of other benefits, such as access to large deposits of rare Earth metals and other materials. The moon, being the closest of these space bodies, is one of the obvious first targets, but humanity still needs some robotic assistance to reap the maximum benefits. So who exactly are these assistants?

The uses of AI

When you want to build a lunar civilization, it helps to know where best to land. Lunar craters are juicy targets, but there is one problem: there are a lot of craters but there is still more bare surface. Picking out a promising crater that contains things like helium-3, uranium and frozen water, is like picking out a needle in a moon-sized haystack, a task of unimaginable trial and error... thus perfect for A.I. to step in. Research from institutions like the Chinese University of Hong Kong, have made A.I. that can scan quickly the area for craters. Compared to manually scanning, this has a comparatively enormous success rate of 83.7 per cent.

Of course, these aren't without difficulties. Firstly, there weren't any pre-set training data to teach the A.I. so the team was forced to construct their own ones

using a different dataset showing a surface crack. Another difficulty was to separate the craters from the rilles, a task that required a pixel-to-pixel analysis. However, despite this feat never being achieved before, the A.I. came out with flying colours, another mark in the race for space.



credits

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THANKS FOR READING!